Norway

SLEIPNER
Volve F
F-1

F-1 A

Plan: 15/9-F-1 A rev0

Standard Survey Report

11 april, 2018

Statoil
Survey Report

Company:	Norway	**Local Co-ordinate Reference:**	Site Volve F
Project:	SLEIPNER	**TVD Reference:**	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	**MD Reference:**	Rotary Table @ 54.90m (Rotary Table)
Well:	F-1	**North Reference:**	Grid
Wellbore:	F-1 A	**Survey Calculation Method:**	Minimum Curvature
Design:	15/9-F-1 A rev0	**Database:**	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	**System Datum:**	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9				
Site Position:		**Northing:**	6,478,563.53 m	**Latitude:**	58° 26' 29.8068 N
From:	Map	**Easting:**	435,050.02 m	**Longitude:**	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	**Slot Radius:**	13.200 in	**Grid Convergence:**	-0.95 °

Well	F-1					
Well Position	**+N/-S**	3.17 m	**Northing:**	6,478,566.69 m	**Latitude:**	58° 26' 29.9072 N
	+E/-W	-3.53 m	**Easting:**	435,046.49 m	**Longitude:**	1° 53' 14.7075 E
Position Uncertainty		0.00 m	**Wellhead Depth:**	91.00 m	**Water Depth:**	91.00 m

Wellbore F-1 A

Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	3NETICREFERENCE	18.08.2013	-1.14	71.62	50,520

Design 15/9-F-1 A rev0

Audit Notes:

Version: **Phase:** PLAN **Tie On Depth:** 2,587.00

Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145.90	3.17	-3.53	333.09

Survey Tool Program **Date** 14.11.2014

From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
150.15	210.70	15/9 F1 36" MWD (F-1)	Magnetic, old	Magnetic Tools (MWD, EMS)
237.60	358.50	15/9 F1 26" Gyro Scientific Modular (F-1)	NOT_useMWD gyro, Baker H	Baker Hughes' MWD gyro service (GyroTrak)
378.00	737.40	15/9 F-1 26" SDI Keeper Gyro Single Shot	Gyro, old	Other Gyro Tools
756.80	1,335.60	15/9 F-1 26" OnTrak (F-1)	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)
1,387.10	2,587.00	15/9 F-1 17 1/2" AutoTrak G3 (F-1)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
2,587.00	3,829.03	15/9-F-1 A rev0 (F-1 A)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)

Planned Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
2,587.00	30.19	33.06	2,443.70	538.48	177.02	395.61	0.000	0.000	0.000
2,610.00	30.19	33.06	2,463.58	548.17	183.33	401.39	0.000	0.000	0.000
2,620.00	29.41	31.13	2,472.26	552.38	185.97	403.95	3.720	-2.349	-5.804
3,224.86	64.18	284.83	2,936.01	783.31	-32.80	708.90	3.730	1.725	-5.272
3,233.11	64.18	284.83	2,939.61	785.21	-39.98	713.84	0.000	0.000	0.000
3,393.60	44.23	285.25	3,033.00	818.76	-165.08	800.38	3.730	-3.729	0.079

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Well:	F-1	North Reference:	Grid
Wellbore:	F-1 A	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-1 A rev0	Database:	Production EDM P246N

Planned Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
3,508.04	44.23	285.25	3,115.00	839.76	-242.09	853.96	0.000	0.000	0.000
3,829.03	44.23	285.25	3,345.00	898.65	-458.11	1,004.26	0.000	0.000	0.000

Design Targets

Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
F-1 A BCU Guide - plan hits target center - Point	0.00	0.00	3,033.00	818.76	-165.08	6,479,382.00	434,885.00	58° 26' 56.1762 N	1° 53' 3.9176 E
F-1 A Top Hugin Target - plan misses target center by 0.02m at 3508.04m MD (3115.00 TVD, 839.76 N, -242.09 E) - Point	0.00	360.00	3,115.00	839.77	-242.11	6,479,403.00	434,808.00	58° 26' 56.8137 N	1° 52' 59.1480 E

Casing Points

Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)
220.90	220.90	30"	30.000	36.000
1,348.40	1,336.38	20"	20.000	26.000
2,596.00	2,451.48	13 3/8"	13.375	17.500

Checked By: ______________________ Approved By: ______________________ Date: ______________